
Radial Velocity and Size of Scrape-Off-Layer Filaments Across 294 MAST Discharges from 100 kHz Visible Imaging

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Abstract

Filaments carry much of the cross-field transport in the tokamak scrape-off layer, and analytic models predict how their radial velocity scales with their size, two quantities measured here on MAST from the 100 kHz, 48×256 pixel strip recordings of the midplane fast camera in the public FAIR-MAST archive. An automatic pipeline built from standard computer-vision operations, comprising a running-minimum background, robust thresholding, connected components, Hungarian linking, least-squares tracks, and optical flow, produces 86,042 filament tracks in 294 discharges across four campaigns. Because the archive distributes no spatial calibration for these recordings, the pixel scale of each camera configuration is recovered by regressing the image column of the plasma limb on the EFIT outboard LCFS radius across discharges, with bootstrap intervals, and lies between 2.7 and 5.2 mm per pixel. The median radial velocity is 424 m/s and the median radial width is 3.6 cm, and after correction for exposure blur the velocity–width relation has a log–log slope of +0.05 (95% interval +0.03 to +0.07), so the velocity is nearly independent of size over the measured range. In two-region units the filaments span $\hat{a} = 0.13$ –1.97 and straddle the velocity maximum of the model, which accounts for the flat dependence, although their normalized velocity of $\hat{v} \approx 0.06$ lies far below the model scale.

1 Introduction

Filaments are field-aligned structures of excess density and temperature that form near the last closed flux surface (LCFS) of a tokamak and move radially outward through the scrape-off layer (SOL), carrying a large part of the cross-field particle and heat flux in the SOL (Krasheninnikov, 2001; Zweben et al., 2007; D’Ippolito et al., 2011). Their radial velocity sets how far they travel before they drain to the divertor targets and thereby the load on the first wall, and analytic models give this velocity as a function of filament size and of the parameters that control how the polarization charge of the filament is dissipated. When the charge drains through the sheath at the target, the velocity falls with size as $v \propto a^{-2}$ (Krasheninnikov, 2001), and when ion polarization current closes the circuit inside the filament, the velocity rises with size as $v \propto a^{1/2}$ (Garcia et al., 2006). The two-region model of Myra et al. (2006) joins these limits and predicts a velocity maximum at a critical size a^* set by the sound speed, the connection length, and the collisionality.

Testing these predictions requires velocities and sizes for many filaments under known plasma conditions, and on the Mega Amp Spherical Tokamak (MAST) the main instrument for this purpose has been a Photron SA1.1 fast camera that views the outboard midplane in visible light. Earlier MAST studies used wide-angle views, identified filaments by hand or with detection software built for that view, and worked from a small number of discharges (Dudson et al., 2008; Ben Ayed et al., 2009; Kirk et al., 2016; Farley et al., 2019), each requiring a camera calibration model fitted to the view in use.

The FAIR-MAST archive (Jackson et al., 2024) makes the full MAST data set public and includes a recording mode of the midplane camera that earlier work did not use at scale, a 48×256 pixel strip through the outboard edge recorded at 100,000 frames per second.

The strip resolves radial motion at $10 \mu\text{s}$ intervals but spans only 48 rows, so it constrains radial motion and radial size and little else, and because the archive includes no spatial calibration for these recordings, the pixel scale has to be recovered from the data.

Here those recordings are turned into filament statistics with an automatic pipeline built from standard computer-vision operations, namely a running-minimum background model, robust thresholding, connected-component detection, Hungarian frame-to-frame assignment, least-squares track fitting, and Lucas–Kanade optical flow as an independent velocity estimate. The pixel scale is calibrated against the equilibrium reconstruction, since the image column of the plasma limb moves with the EFIT outboard LCFS radius from shot to shot. A robust regression over a campaign then gives the scale in millimetres per pixel with a bootstrap confidence interval for each camera configuration, using only the images and EFIT, and the pipeline is run on every usable strip recording in the archive.

Applied to the archive, the pipeline yields radial velocity and radial width for 86,042 filament tracks in 294 MAST discharges, with a median radial velocity of 424 m/s, a median radial full width at half maximum of 3.6 cm, and 81% of tracks moving outward. The pooled velocity–size slope is $+0.09$ in log–log units and $+0.05$ after the widths are corrected for exposure blur, so the velocity is nearly independent of size over the measured range, and in two-region units the filaments sit at $\hat{a} \approx 0.13\text{--}1.97$, where the model places its velocity maximum, although their normalized velocity $\hat{v} \approx 0.06$ is far below the model scale. Robust regression gives the dependences of the per-shot median velocity and width on plasma current, Greenwald fraction, and beam power with confidence intervals, and code and derived data accompany the paper.

2 Background

Filament motion. A filament is a field-aligned tube of plasma, with a cross-field radius a between a few ion gyroradii and a few centimetres, that is denser and hotter than the surrounding SOL. In the outboard SOL the magnetic curvature and gradient drifts separate ions and electrons across the filament, and the resulting electric field drives an $E \times B$ drift that moves the whole filament radially outward (Krasheninnikov, 2001; D’Ippolito et al., 2011). The drift speed depends on the potential that the charge separation can build, and therefore on the paths through which the polarization charge is dissipated, which leads to two standard limits. If the filament is connected along the field line to the divertor target, the charge drains through the sheath and $v \propto c_s (L_{\parallel}/R) (\rho_s/a)^2$, where c_s is the sound speed, ρ_s the ion sound gyroradius, $L_{\parallel}$ the connection length, and R the major radius (Krasheninnikov, 2001), whereas if the ion polarization current closes the circuit inside the filament, $v \propto c_s (a/R)^{1/2}$ (Garcia et al., 2006).

The two-region model. Myra et al. (2006) model the midplane and the divertor leg as two coupled regions and join these limits. With

$$a^* = \rho_s^{4/5} L_{\parallel}^{2/5} R^{-1/5}, \quad v^* = c_s \left(\frac{2a^*}{R} \right)^{1/2}, \quad \Theta = \left(\frac{a}{a^*} \right)^{5/2}, \quad \Lambda = \frac{v_{ei} L_{\parallel}}{\Omega_e \rho_s}, \quad (1)$$

and $\hat{a} = a/a^*$, $\hat{v} = v/v^*$, the model gives

$$\hat{v} = \begin{cases} \hat{a}^{1/2} & \text{connected ideal interchange } (\Lambda < 1, \Theta < 1), \\ \hat{a}^{-2} & \text{sheath connected } (\Lambda < 1, \Theta > 1), \\ \Lambda \hat{a}^{-2} & \text{resistive X-point } (1 < \Lambda < \Theta), \\ \hat{a}^{1/2} & \text{resistive ballooning } (\Lambda > 1, \Lambda > \Theta). \end{cases} \quad (2)$$

Here v_{ei} is the electron–ion collision frequency and Ω_e the electron gyrofrequency (D’Ippolito et al., 2011), and at low collisionality the velocity rises as $\hat{a}^{1/2}$ for small filaments, peaks near $\hat{a} = 1$, and falls as $\hat{a}^{-2}$ for large ones. The model is a scaling argument whose prefactors depend on the filament profile and on the parallel structure, and simulations in realistic geometry find velocities below the model values by factors of order unity

(Easy et al., 2014; Walkden et al., 2015; Paruta et al., 2018). Theiler et al. (2009) measured the velocity–size relation on the TORPEX basic plasma device and recovered the predicted transitions between regimes.

Camera measurements. Visible-light imaging records line-integrated D_α emission, which depends on the local density, temperature, and neutral density. Filaments appear as bright, field-aligned streaks against the quiescent background, and their emission tracks the density perturbation closely enough that velocities and widths from imaging agree with probe measurements within tens of percent (Zweben et al., 2007; Ben Ayed et al., 2009). In a tangential view the brightest image feature is the plasma limb, the band where the line of sight is tangent to the flux surfaces near the LCFS, and filaments cross the limb as they move outward. Because the camera integrates over each exposure, a filament that moves a distance $v t_{\text{exp}}$ during the exposure is smeared by that distance along its direction of motion, an effect treated explicitly in Section 6.

3 Data

Archive. FAIR-MAST (Jackson et al., 2024) serves the MAST data set, which covers 15,969 discharges, as one Zarr store per discharge over HTTP, with a metadata API for discovering signals and shot parameters. Six groups are read from it, namely the midplane fast camera (rbb), the EFIT equilibrium reconstruction (efm), the summary signals for line-averaged density and Greenwald density (esm), the interferometer (ane), the core and edge Thomson scattering systems (ayc, aye), and the midplane D_α monitor (xim), while plasma current, heating scheme, injected beam power, and the flat-top interval come from the per-shot metadata table.

Camera. The rbb signal is the Photron SA1.1 fast camera (“bullet camera B”) at the HM10 midplane port, which views the outboard edge tangentially (Kirk et al., 2016; Farley et al., 2019), and in the recording mode used here it reads out a 48×256 pixel region of its sensor at 100,000 frames per second with 10-bit depth, with the offset of the region on the sensor recorded with each shot. Exposure times are 1–10 μs , with 4, 7, and 10 μs the common settings, and the view string in the metadata records the lens, aperture, and filter in use and whether the camera was mounted inverted, although the SOL-side detection in Section 4 makes the pipeline independent of this orientation. Appendix A lists the camera configurations that these settings define in the processed set, together with their calibration.

Selection. Of the archive, 468 discharges have a 48×256 rbb recording, each a burst at the fast frame rate embedded in a slow record of the whole discharge at 1–2 kHz, and the burst lasts 100 ms in the median (range 3–500 ms), with a frame interval of 10 μs in 419 recordings. After the time base of every recording is read, a discharge is kept if its burst overlaps the plasma flat top, extended by 20 ms on each side, by at least 2,000 fast frames (375 discharges), if its flat top is defined and at least 50 ms long (404), and if its peak plasma current is at least 400 kA (421), and 368 discharges satisfy all three conditions. The burst frames inside the extended flat top were downloaded and processed for 368 discharges, with a median window of 186 ms and 3.5 million frames in total, and the 16 of these with a burst frame rate of 50 kHz are processed but kept out of the statistics so that the time resolution is uniform. Table 1 summarizes the processed set by campaign, with the number of discharges, the exposure settings, the pixel scales, and the median velocity and width for each.

Equilibrium and plasma parameters. From EFIT, reconstructed every 5 ms, the LCFS boundary contour, q_{95} , the plasma current, the magnetic axis position, the vacuum toroidal field at the axis, and the minor radius are taken at each time, and the outboard LCFS radius is the largest major radius of the boundary contour within 5 cm of the axis height, which is the quantity that the camera limb tracks. Line-averaged density comes from the summary signal where it exists and from the CO₂ interferometer summary otherwise, and the Greenwald fraction uses $n_{\text{GW}} = I_p / (\pi a^2)$ with I_p in MA and a the EFIT minor radius. Thomson

Campaign	Years	Shots	In stats	Exp. (μs)	Pooled scale (mm/px)	Own scales	$\bar{v}_r$ (m/s)	$\bar{\delta}_r$ (cm)
M6	2006–2007	257	220	1, 5, 7, 10	3.7 (3.3–4.1)	3 (2.9–3.7)	419	3.6
M7	2008–2009	15	13	4, 10	3.3 (3.0–3.8)	1 (3.3)	451	3.6
M8	2011	32	25	2, 4	5.2 (3.6–11.5)	1 (5.2)	627	5.2
M9	2013	48	36	4, 10	2.9 (2.5–4.2)	1 (2.7)	323	2.8

Table 1: Processed discharges by campaign, where “In stats” counts discharges with at least 50 selected tracks, the pooled and own scales are the calibrations of Section 4.6 with their 95% bootstrap intervals, and $\bar{v}_r$ and $\bar{\delta}_r$ are the medians over discharges of the per-discharge median radial velocity and width.

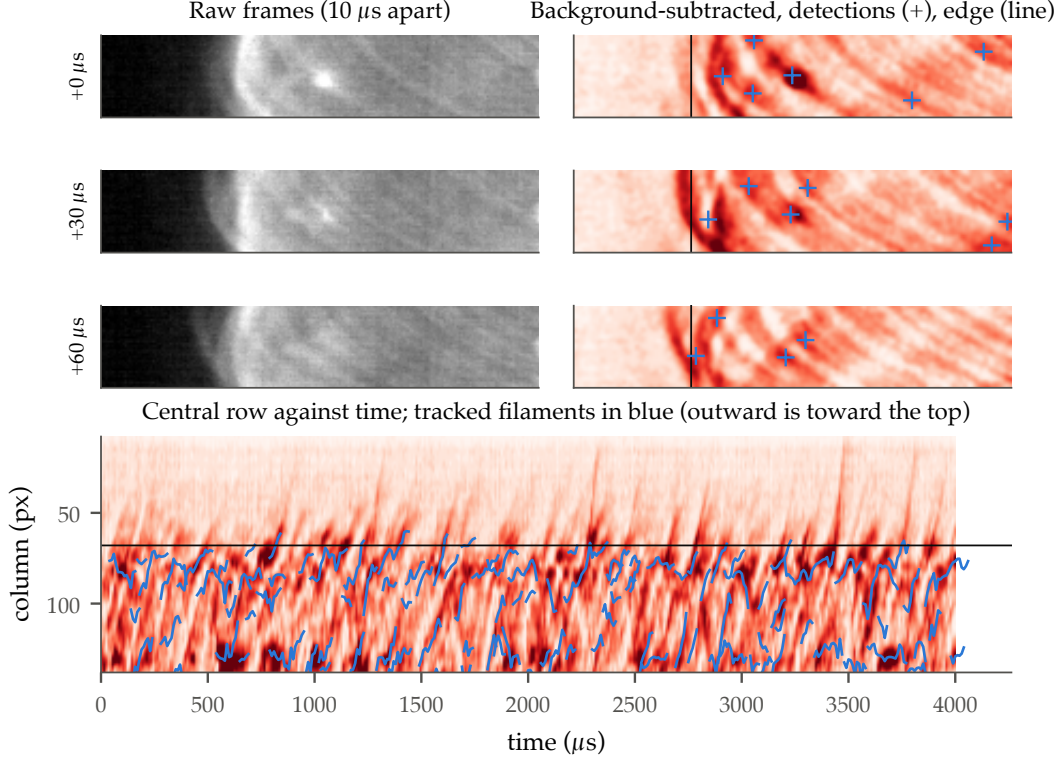


Figure 1: Pipeline stages on discharge 15235 (M6, 7 μs exposure), with three raw frames 30 μs apart and the corresponding background-subtracted frames with detections (crosses) and the edge column (line) in the top row, and the central row of the background-subtracted strip against time with tracked filaments overlaid in the bottom row, where filaments appear as inclined streaks that move toward the SOL (upward in this view).

profiles exist for the later campaigns only, and a profile reaches the LCFS during the flat top for 60 of the 294 discharges in the statistics.

4 Method

Figure 1 shows the stages of the pipeline on one discharge, and every stage runs with the same settings on every shot without any per-shot tuning.

4.1 Strip geometry and the plasma edge

A recording is an array of n frames of 48×256 pixels with 10-bit depth, whose columns run across the plasma edge so that the SOL occupies one end of the strip and the core the

other, with the end depending on how the camera was mounted in that campaign. The row-averaged brightness profile $p(t, x)$ is smoothed over 200 frames (2 ms) in time and $\sigma = 2$ columns in space, and the darker outer quarter of the columns is taken as the SOL side, which fixes a sign $s = \pm 1$ such that outward motion is $s dx/dt$. The edge column $x_{\text{half}}(t)$ is the first column, scanning from the SOL side, at which p exceeds the midpoint between the dark level, defined as the mean of the ten outermost SOL-side columns, and the interior level, defined as the median over the columns 40–75% of the way across the strip from the SOL side, with linear sub-pixel interpolation. The limb peak $x_{\text{peak}}(t)$ is the maximum of p within 40 columns inboard of x_{half} , refined by parabolic interpolation, and detections are referenced to x_{half} during processing and shifted to x_{peak} at aggregation, because the peak is the feature that the calibration in Section 4.6 ties to the equilibrium.

4.2 Background subtraction and detection

Because filaments are transient, a per-pixel running minimum over 21 frames (± 10 frames, $\pm 100 \mu\text{s}$) estimates the quiescent background (Kirk et al., 2016), and the fluctuation image is the frame minus this background, passed through a 3×3 median filter and a Gaussian with $\sigma = 1$ pixel. The noise level σ_n is 1.4826 times the median absolute deviation of all fluctuation pixels in a block of 3,000 frames, pixels above $3\sigma_n$ are labelled into 8-connected components, and components smaller than 12 pixels are dropped. Each remaining component yields its intensity-weighted centroid (x_c, y_c) , its peak fluctuation, its orientation and major and minor axis lengths from its second central moments, and its radial width, which is measured on a chord as the number of columns at or above half the maximum of the fluctuation averaged over the five rows centred on y_c . The width is measured on a chord because filaments appear as inclined streaks and the column extent of a whole component grows with the tilt, and the signed SOL distance of a detection is $s(x_c - x_{\text{half}})$.

4.3 Tracking

Detections in consecutive frames are linked by the Hungarian method (Kuhn, 1955), with the Euclidean distance between centroids as the cost and a gate of 12 pixels per frame, which corresponds to about 4.5 cm or 4.5 km/s at the calibrated scale. Unmatched detections start new tracks, a track ends at the first frame without a match, and tracks with fewer than five detections ($50 \mu\text{s}$) are discarded. The radial velocity of a track in pixels per frame is the least-squares slope of $s x_c$ against frame index, with a standard error from the residuals, while its width is the median chord FWHM over the track and its SOL distance the median signed distance. Recordings are processed in blocks of 3,000 frames with a 60-frame overlap, and tracks that start in the overlap are dropped from the earlier block and recovered in the later one.

4.4 Optical flow as an independent velocity

As a check on the tracking, the velocity at each detection is also estimated by Lucas–Kanade optical flow (Lucas & Kanade, 1981), in which Sobel spatial gradients and the frame difference in a 7×7 window around the centroid give a 2×2 least-squares system for the displacement between frames f and $f + 1$, solved when its condition number is below 10^4 . The flow velocity of a track is the mean of the horizontal component over its detections with the sign s applied, and because this estimate uses no assignment step and responds to the whole brightness profile around the centroid, agreement between the two estimates is evidence that neither is an artefact of its method.

4.5 ELM detection

Edge-localized modes (ELMs) eject large filaments that would dominate a velocity distribution, so they are detected from the midplane D_α monitor, sampled at 50–250 kHz depending on the campaign, as spikes where the residual after a 5 ms running median exceeds six times its robust standard deviation, with the time of each spike taken as the centre of mass

of its samples. In some campaigns the monitor is coarsely digitized, with only tens of distinct values, which makes the robust standard deviation vanish, so the threshold is floored at six digitizer steps and at half the inter-ELM level. Because the D_α channel is absent or saturated in some discharges, the same detector is also applied to the frame-mean camera brightness, and a track that begins within 1 ms of a spike of either kind is flagged, while discharges with more than five D_α spikes inside the camera window are labelled ELMy. Flagged tracks are excluded from all statistics, and the parameter dependences in Section 6 use the non-ELMy discharges only.

4.6 Pixel scale from the equilibrium

The archive gives no spatial calibration for the strip recordings, and the scale is therefore recovered from the data. Because the limb peak sits where the line of sight is tangent to the flux surfaces at the LCFS, its sensor column should move with the outboard LCFS radius R_{LCFS} from EFIT, and the slope of column on radius is the inverse pixel scale. Within a single discharge R_{LCFS} moves by only a few centimetres during the flat top, the EFIT reconstruction is least reliable during ramp-up and termination, and the brightness profile changes shape as the density rises, so a within-shot regression is unstable (Appendix B). Between discharges, by contrast, the plasma position varies by 10–20 cm while the camera and its optics stay fixed through a campaign, which gives a much longer lever arm.

For each discharge, R_{LCFS} is taken at every EFIT time inside the flat top as the largest major radius of the EFIT boundary contour within 5 cm of the magnetic-axis height, x_{peak} is taken from the profile smoothed over 2 ms, and EFIT samples whose R_{LCFS} differs by more than 8 cm from the flat-top median are dropped as unconverged. Each discharge then contributes one point, its median x_{peak} , converted to sensor coordinates by adding the region-of-interest offset recorded in the camera metadata, against its median R_{LCFS} . A camera configuration is defined as the combination of campaign, view string in the camera metadata, and region-of-interest offset, because a change in any of these can mean a change of lens, pointing, or sensor region. A configuration with at least 8 discharges and at least 3 cm of spread in R_{LCFS} receives its own Theil–Sen slope (Sen, 1968), which is kept if its sign agrees with s , the Spearman correlation satisfies $|\rho| \geq 0.5$, and the bootstrap interval excludes zero. Every other configuration uses the pooled slope of its campaign, fitted on all configurations of the campaign after the mean R_{LCFS} and mean column of each configuration are subtracted to remove the per-configuration intercepts, and in all cases the 95% intervals come from 2,000 bootstrap resamples of discharges within configurations (Efron & Tibshirani, 1993) and the scale is the inverse of the slope magnitude.

4.7 Physical units and track selection

The radial velocity of a track is $v_r = s \dot{x}_c \times f_{\text{frame}} \times \ell$, where ℓ is the pixel scale, and its radial width is $\delta_r = \text{FWHM} \times \ell$, while the SOL distance is measured from x_{peak} using the per-shot median offset $x_{\text{peak}} - x_{\text{half}}$. A track enters the statistics if its median position lies between 10 pixels inboard and 40 pixels outboard of the limb peak (3–5 cm inboard to 11–21 cm outboard, depending on the scale of the configuration), its velocity standard error is below 0.5 pixels per frame, its width is finite, it begins inside the plasma flat top, and it is free of an ELM flag. A discharge enters the shot-level statistics if it has at least 50 such tracks with $v_r > 0$, which keeps the per-discharge medians stable.

4.8 Statistics

All central values are medians with 95% bootstrap intervals from 2,000 resamples, and slopes are Theil–Sen estimates in $\log_{10}$ – $\log_{10}$ coordinates with the 95% interval of the estimator, reported together with Spearman’s ρ . The velocity–size slope is computed per discharge and also pooled, and for the pooled fit the v_r and δ_r of each track are divided by the medians of its discharge so that differences between discharges in the pixel scale or the plasma conditions do not create a spurious correlation. The correction for exposure blur replaces δ_r by $\delta_{\text{corr}} = (\delta_r^2 - (v_r t_{\text{exp}})^2)^{1/2}$, which treats the smear as a Gaussian

of width $v_r t_{\text{exp}}$ added in quadrature, before the slope is refitted. Parameter dependences are Theil–Sen slopes of the per-discharge median v_r and δ_r on plasma current, Greenwald fraction, and injected beam power, with the beam power offset by 0.05 MW so that Ohmic discharges enter the log fit.

The two-region normalization requires T_e and n_e at the LCFS, B , q_{95} , and R , and the temperature comes from the Thomson scattering profile, from the edge system where it exists and the core system otherwise, interpolated at R_{LCFS} and taken as the median over the flat top, with $T_e = 30$ eV assumed and marked as such when no profile reaches the LCFS. The separatrix density $n_{e,\text{sep}}$ is taken as half the line-averaged density, B as the vacuum toroidal field at the magnetic axis scaled by $R_{\text{axis}}/R_{\text{LCFS}}$, and the connection length as $L_{\parallel} = \pi q_{95} R_{\text{axis}}$, while the filament radius in the model is half the measured FWHM, $a = \delta_r/2$, and v_{ei} follows the NRL formulary (Huba, 2019) with deuterium ions.

5 Results

5.1 Pixel scale

Of the 23 camera configurations listed in Table 2 in the appendix, 6 have enough discharges and enough spread in R_{LCFS} for their own fit and cover 241 of the 352 processed discharges, while the rest use the pooled slope of their campaign (Figure 2). The limb-peak column falls linearly with R_{LCFS} in every fitted configuration, with Spearman correlations between -0.58 and -0.94 , the sign of every slope agrees with the SOL direction found from the images, and the fitted scales span 2.7–5.2 mm per pixel. The tightest fit is the M6 configuration “HM10, image normal, f/1.8” (70 discharges, $\rho = -0.94$), at 2.9 mm per pixel (2.7–3.0), and the M9 configuration gives 2.7 mm per pixel (2.5–3.4). For the same camera, port, and campaign, Farley et al. (2019) quote a resolution of 5 mm at the tangency plane for wide-angle images reduced to 256×160 pixels from the 640×448 mode of the camera, which is about 2 mm per native pixel, so the M9 value is of that order, although 2 mm lies below its interval and the archive does not record whether the lens was the same.

Two features of the calibration limit what the physical units can support, the first being that the intervals are 10–30% wide and cannot capture a bias common to a whole configuration, such as a drift of the emission peak relative to the LCFS with density. The second is that the filaments look alike in pixel units in every configuration, since across the 6 fitted configurations with at least ten discharges the median track velocity is 1.15–1.40 pixels per frame and the median width is 8–13 pixels, while the fitted scales differ by a factor of 1.9. The differences between campaigns in Table 1 therefore reflect differences between their calibrations and are left uninterpreted. Because a scale error multiplies v_r and δ_r of a configuration by the same factor, it cancels in the velocity–size slopes of Section 6, which are fitted after each discharge is normalized by its own medians, and it moves a discharge along a diagonal in the regime diagram without changing the conclusions drawn there.

5.2 Velocity and width distributions

The statistics include the 294 discharges with at least 50 selected tracks, 86,042 tracks in total, and Figure 3 shows the pooled distributions, with a median radial velocity over discharges of 424 m/s (95% interval 406–439) and a median radial width of 3.6 cm (3.4–3.7). The velocity distribution is skewed with a tail to 2–3 km/s, and the width distribution peaks at 2–3 cm with a tail to 10 cm. Among all near-edge tracks that pass the quality cuts, 81% move outward, and the inward-moving remainder is concentrated inboard of the limb peak, where the field-aligned streaks sweep across the strip as they rotate with the plasma, so only outward-moving tracks enter the velocity–size analysis.

5.3 Independent velocity estimates

Two estimates that do not use the tracking confirm the velocity scale, the first being the cross-correlation lag between the row-averaged fluctuation profiles of consecutive frames inside the near-edge window, which involves no detection step and gives a per-discharge

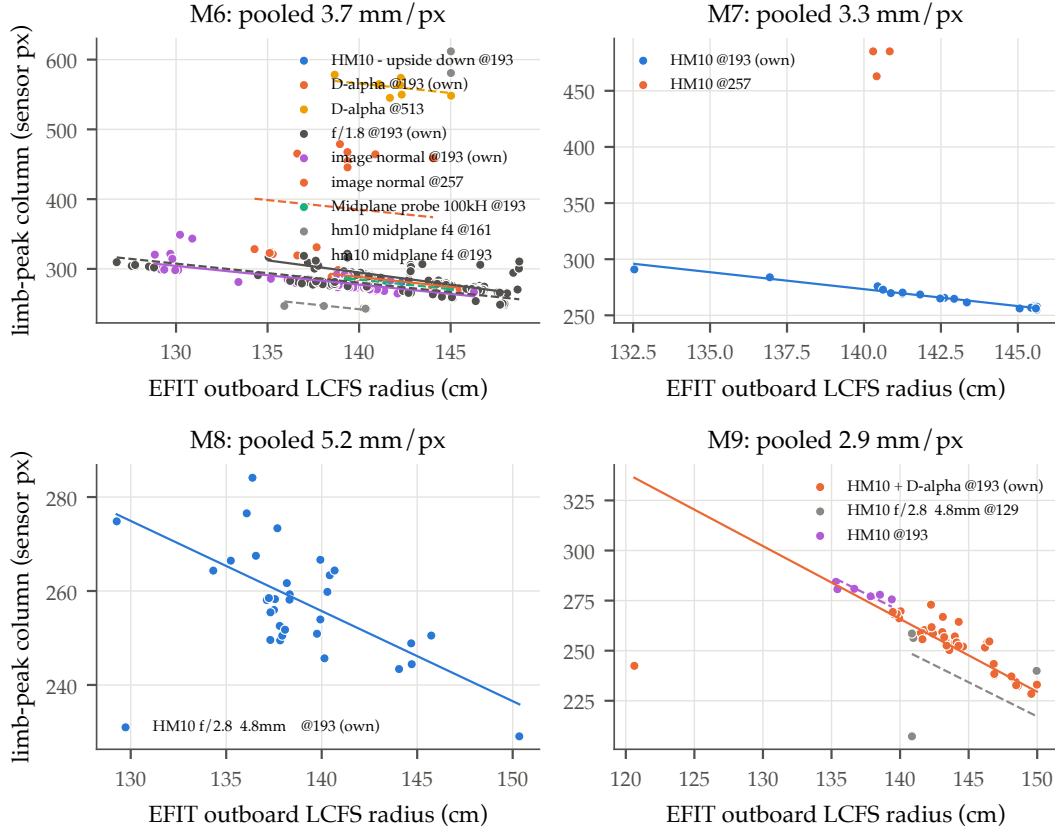


Figure 2: Between-shot pixel-scale calibration, in which each point is one discharge, placed at its median limb-peak column in sensor coordinates over the flat top against its median EFIT outboard LCFS radius. Colours mark camera configurations (view string and region-of-interest offset), solid lines show the Theil-Sen fit of each configuration, dashed lines show the pooled campaign slope with the intercept of the configuration, and the scale is the inverse of the slope.

median velocity of 257 m/s over the 175 discharges in which more than half the frame pairs have a usable correlation peak. Its ratio to the tracking median is 0.63 (0.52–0.68), and the two correlate across discharges with Spearman $\rho = +0.29$, while a ratio below one is expected because the lag method weights every bright structure in the window, including the slow inboard part of the streaks and the stationary limb. The second, Lucas-Kanade flow at the detections, correlates with the track velocities within discharges, with a median per-discharge correlation of +0.23 and $\rho = +0.49$ between discharge medians, but is smaller by a factor of 0.13. Because the streaks are elongated and inclined, the brightness-constancy constraint fixes only the component of motion normal to the streak, and displacements exceed one pixel per frame, both of which bias the flow magnitude down, so the flow serves only as a check on sign and ordering.

6 Analysis

6.1 Velocity against size

Figure 4 shows the velocity-size relation after each discharge is normalized by its own medians, and the per-discharge Theil-Sen slope of $\log v_r$ on $\log \delta_r$ has a median of +0.11 (+0.08–+0.15) and is positive in 70% of discharges, while the pooled slope is +0.09 (+0.06–

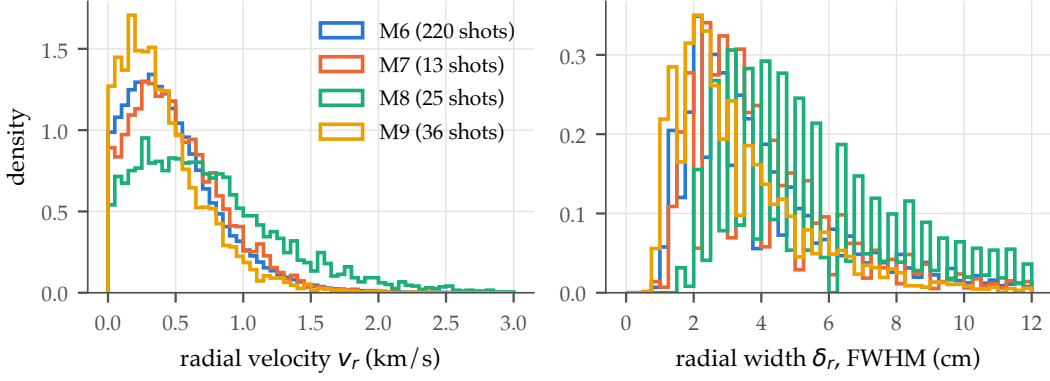


Figure 3: Distributions of radial velocity (left) and radial FWHM (right) of the selected tracks in each campaign, pooled over the discharges in the statistics.

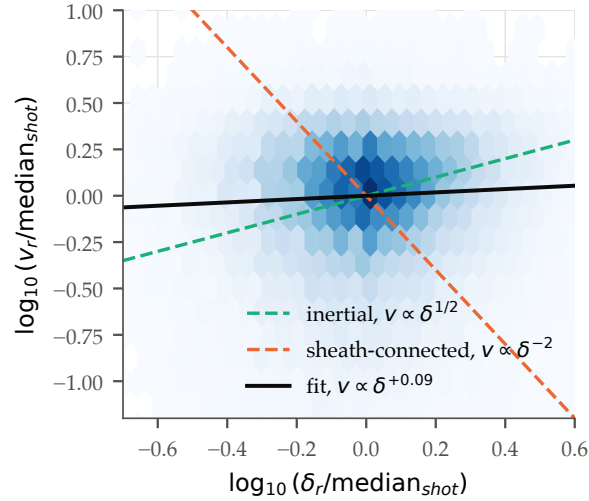


Figure 4: Radial velocity against radial width for all selected tracks, each normalized by the median of its discharge and shown as hexagonal bin density, with the pooled Theil–Sen fit as a solid line and the sheath-connected and inertial scalings through the medians as dashed lines.

+0.11) with Spearman $\rho = +0.05$. The relation is far from both limiting scalings, since the interval excludes the sheath-connected δ^{-2} and lies well below the inertial $\delta^{1/2}$.

6.2 Exposure blur

A filament that moves during the exposure is smeared along its direction of motion, and because faster filaments are smeared more, the smear produces a positive velocity–width correlation with no physical content. The median smear $v_r t_{\text{exp}}$ is 0.31 cm, about a tenth of the median width, so the effect is small but measurable, and two tests bound it. In the first test, the slope depends on exposure in the expected direction within M6, where the two common settings share the optics, with a median per-discharge slope of +0.05 (+0.01–+0.11) at 7 μs ($n = 105$) and +0.16 (+0.11–+0.27) at 10 μs ($n = 142$), while the median width stays the same, 3.4 against 3.4 cm, because the extra smear of 3 μs at 400 m/s is 0.1 cm. The 4 μs discharges belong to M7 and M8 and cannot be compared in width with those of M6, although their slope, +0.07 (−0.04–+0.16), is also low. In the second test, correcting each width for its own smear in quadrature lowers the pooled slope to +0.05 (+0.03–+0.07),

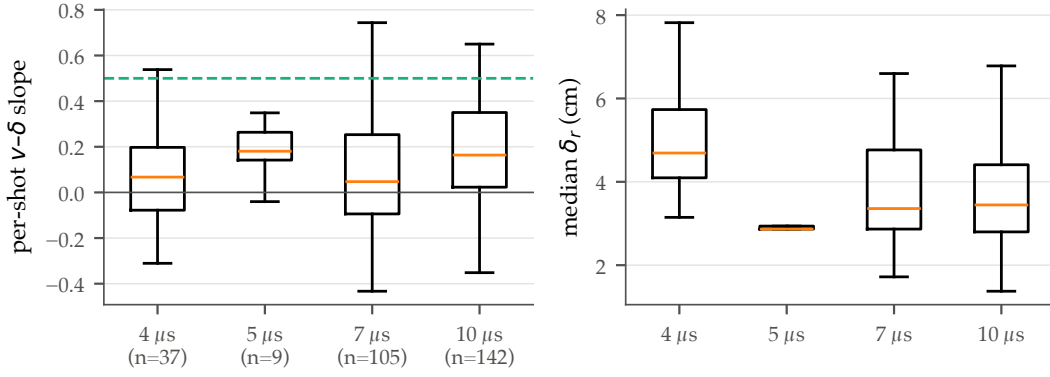


Figure 5: Per-discharge velocity–width slopes by exposure setting (left), with the inertial scaling as a dashed line, and per-discharge median width by exposure setting (right).

and this corrected value is taken as the measurement, which shows that the radial velocity is nearly independent of width over the measured range, with at most a weak positive dependence.

6.3 Two-region normalization

With the inputs of Section 4, the median reference scales are $a^* = 2.6$ cm, $v^* = 7345$ m/s, and $\Lambda = 0.3$, which puts the discharges in the low-collisionality half of the regime diagram, and Figure 6 places each discharge at its median $\hat{a} = (\delta_r/2)/a^*$ and $\hat{v} = v_r/v^*$. The filaments span $\hat{a} = 0.13\text{--}1.97$ (median 0.69), a range that straddles the velocity maximum of the model and over whose central half the model $\hat{v}(\hat{a})$ stays within a factor of two of its peak, so a weak net dependence of velocity on size is expected, and this is what the data show. The normalized velocity, however, is only $\hat{v} \approx 0.06$, which means that the measured filaments move at a few percent of the velocity scale of the model. Part of this gap comes from the input assumptions, since $v^* \propto T_e^{7/10}$ and T_e at the LCFS is assumed for 234 of the 294 discharges, and part from the model itself, which gives the velocity of an isolated, fully formed filament with an order-unity prefactor that simulations in realistic geometry find to be below one (Walkden et al., 2015; Paruta et al., 2018). The measurement is also made within 15 cm of the limb, where filaments are still forming and accelerating and may not yet have reached the velocity that the model describes. The ratio is nonetheless larger than the uncertainty in the prefactor alone explains, so the amplitude of the velocity, more than its dependence on size, is the point of tension with the model.

6.4 Dependence on plasma parameters

Figure 7 shows the per-discharge medians against plasma current and Greenwald fraction for the 183 non-ELMy discharges, which span $I_p = 402\text{--}1127$ kA and $f_{\text{GW}} = 0.14\text{--}1.01$, and no dependence on either is resolved. The log–log slope of the median velocity is -0.05 ($-0.29\text{--}+0.19$) on plasma current and -0.06 ($-0.18\text{--}+0.05$) on Greenwald fraction, the corresponding slopes for the width are $+0.10$ ($-0.21\text{--}+0.40$) and $+0.06$ ($-0.08\text{--}+0.21$), and beam power has no effect on the velocity (-0.01 , $-0.03\text{--}+0.01$) and a small positive effect on the width ($+0.05$, $+0.01\text{--}+0.07$). Because these fits pool configurations with different calibrations, they are repeated inside the best-calibrated configuration alone (“HM10, image normal, $f/1.8''$ ”, 37 non-ELMy discharges at one scale, $I_p = 606\text{--}1127$ kA), where the velocity slopes are -0.29 ($-1.02\text{--}+0.66$) on current and -0.10 ($-0.26\text{--}+0.12$) on Greenwald fraction, consistent with the pooled result. Kirk et al. (2016) found in a dedicated current scan at fixed density and power that the radial velocity falls with current, whereas in the present sample current, density, heating, and campaign vary together and most discharges sit near 700 kA,

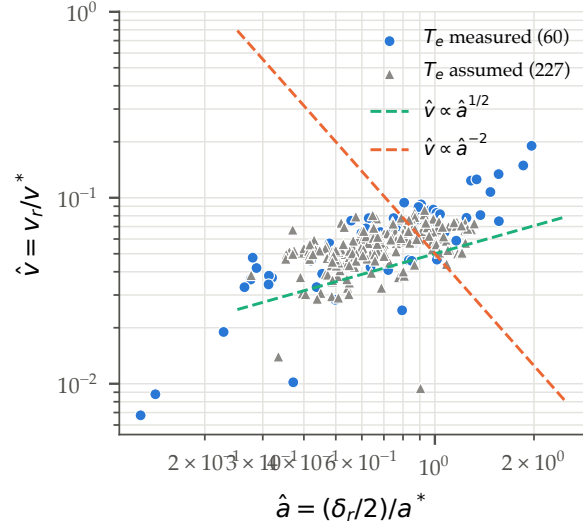


Figure 6: Per-discharge median velocity and size in two-region units, with the inertial ($\hat{v} = \hat{a}^{1/2}$) and sheath-connected ($\hat{v} = \hat{a}^{-2}$) branches as dashed lines, circles for discharges with a measured T_e at the LCFS, and triangles for discharges with an assumed $T_e = 30$ eV.

so confounding could hide a weak dependence, and the slopes are reported as measured without any claim of contradiction.

7 Related Work

Filament imaging on MAST. [Dudson et al. \(2008\)](#) used the midplane fast camera with automated analysis of L-mode images and BOUT simulations and found that the toroidal mode number of the edge turbulence rises with density and falls with q_{95} , that filament width has the opposite dependence, and that simulated filaments are born within about 2 cm of the edge in a self-generated $E \times B$ shear layer. [Ben Ayed et al. \(2009\)](#) combined camera images with a reciprocating Langmuir probe during inter-ELM periods, showed that filaments rotate near the LCFS and move outward, with more and faster filaments at higher density, and used the radial velocities to test the sheath-connected and inertial scalings. [Kirk et al. \(2016\)](#) scanned the plasma current at fixed density and heating power in L-mode and found that the radial velocity, and to a lesser extent the radial size, decrease as the current rises, a dependence consistent with the measured divertor profiles. [Farley et al. \(2019\)](#) introduced a tomographic inversion of wide-angle images onto a basis of equilibrium field lines, which turns filament identification into a problem in field-aligned coordinates, and all of these studies relied on a camera geometry model built with Calcam ([Silburn et al., 2018](#)) for the view in use. The present work departs from them in using the 48-row strip mode at 100 kHz in place of wide-angle frames, in recovering the pixel scale from the equilibrium without a geometry model, and in processing hundreds of discharges across four campaigns where earlier studies scanned a few.

Blob theory and simulation. The sheath-connected scaling is due to [Krasheninnikov \(2001\)](#) and the inertial scaling to [Garcia et al. \(2006\)](#), and [Myra et al. \(2006\)](#) introduced the two-region model used here for normalization, which [D’Ippolito et al. \(2011\)](#) review together with its comparisons with experiment. [Theiler et al. \(2009\)](#) measured the velocity–size relation on TORPEX and recovered the predicted regime transitions, and three-dimensional simulations of isolated filaments ([Easy et al., 2014](#); [Walkden et al., 2015](#); [Paruta et al., 2018](#)) find that the scalings hold in form while realistic geometry and parallel dynamics change the prefactors. [Zweben et al. \(2007\)](#) review edge turbulence imaging across devices, including the relation between light emission and density fluctuations on which imaging measurements rely.

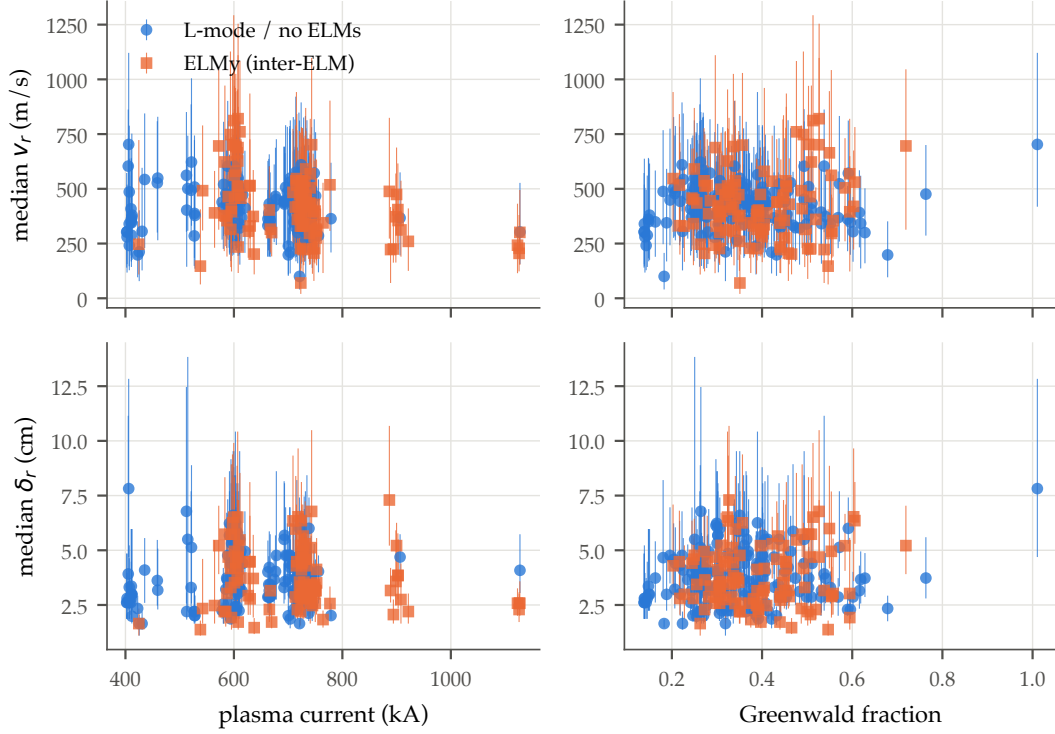


Figure 7: Per-discharge median radial velocity (top) and width (bottom) against plasma current (left) and Greenwald fraction (right), with interquartile ranges over tracks as bars, circles for discharges without ELMs in the camera window, and squares for ELMy discharges measured between ELMs.

Methods. The detection and tracking steps are standard, consisting of connected-component labelling after robust thresholding, frame-to-frame assignment by the Hungarian method (Kuhn, 1955), and Lucas–Kanade optical flow (Lucas & Kanade, 1981), while slopes use the Theil–Sen estimator (Sen, 1968), confidence intervals use the bootstrap (Efron & Tibshirani, 1993), and the data come from the FAIR-MAST archive (Jackson et al., 2024).

8 Limitations

Calibration. The pixel scale rests on the assumption that the limb peak tracks the EFIT outboard LCFS radius with a fixed offset within a camera configuration, but the offset can drift with the emission profile, which depends on density and neutral pressure, and the EFIT boundary carries its own uncertainty of order a centimetre. The bootstrap intervals in Table 2 include the scatter that these effects produce but cannot capture a bias common to a whole configuration. The near-uniformity of the filaments in pixel units across configurations whose fitted scales differ by almost a factor of two (Section 5) is either a coincidence or a sign that part of the spread in the fitted scales has a non-optical origin, which the archive cannot decide, and no conclusion in the paper therefore depends on comparing physical units between configurations. Every velocity and width in the paper carries a systematic uncertainty of 10–30% from the scale of its configuration, and a comparison between configurations carries an uncertainty of up to a factor of two. A per-shot Calcam model would remove this uncertainty and is the natural next step if the calibration images can be released, and Appendix B shows why the within-shot regression cannot substitute for it.

Geometry. The strip gives a single row band through the edge whose column axis is close to radial only at the tangency plane, and the measured width is the chord length of an inclined streak, which overestimates the width perpendicular to the filament by $1/\cos\theta$, where θ is the angle between the streak and the column axis. This factor is left uncorrected because the angle is poorly determined in a 48-row strip, and since it is roughly constant within a campaign, it shifts $\hat{a}$ while leaving the velocity–size slope unchanged. Emission is line-integrated along the view, so a bright structure behind the tangency plane projects to a smaller apparent radius, which blurs the SOL distance while leaving the velocity, a time derivative, unaffected.

What the camera sees. The D_α emission depends on neutral density as well as plasma density, and because the neutral density falls steeply inside the LCFS, filaments are visible mainly in the SOL and change in brightness as they move. Tracking by centroid is robust to a slowly changing amplitude, whereas optical flow is sensitive to it, which is one reason the flow velocities are low.

Selection. The measurement region extends from 10 pixels inboard to 40 pixels outboard of the limb, that is from 3–5 cm inboard to 11–21 cm outboard depending on the configuration, and includes filaments at birth, so velocities in the far SOL may differ. The threshold of 50 tracks per discharge and the exclusion of ELM-adjacent tracks remove some discharges and some of the largest events, so the headline numbers describe inter-ELM and L-mode filaments near the edge.

Model inputs. The LCFS temperature is assumed for most discharges and $n_{e,\text{sep}}$ is taken as half the line average, and because both enter a^* , v^* , and Λ as power laws with exponents below one, a factor of two in either moves a discharge by less than a factor of two in the regime diagram. The conclusion that the sample straddles $\hat{a} = 1$ is robust to this uncertainty, whereas $\hat{v}$ itself is known only to within a factor of two.

9 Conclusion

Public archives make it possible to measure filament statistics on hundreds of discharges with a pipeline that requires no tuning by hand, and on MAST the 100 kHz strip recordings give radial velocities and widths for 86,042 filaments in 294 discharges, with a pixel scale recovered from the equilibrium to 10–30% for each camera configuration. The radial velocity has a median of 424 m/s and, once exposure blur is removed, depends on filament width with a log–log slope no larger than about 0.1. In two-region units the filaments sit around the predicted velocity maximum, which explains the flat dependence, although they move at a few percent of the velocity scale of the model. The amplitude discrepancy, the campaign-level calibration, and the radial-only geometry are the three areas in which a dedicated experiment or a released geometry model would improve on what the archive alone can give.

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References

- N. Ben Ayed, A. Kirk, B. Dudson, S. Tallents, R. G. L. Vann, H. R. Wilson, and the MAST Team. Inter-ELM filaments and turbulent transport in the Mega-Amp Spherical Tokamak. *Plasma Physics and Controlled Fusion*, 51(3):035016, 2009. doi: 10.1088/0741-3335/51/3/035016.

-
- D. A. D'Ippolito, J. R. Myra, and S. J. Zweben. Convective transport by intermittent blob-filaments: Comparison of theory and experiment. *Physics of Plasmas*, 18(6):060501, 2011. doi: 10.1063/1.3594609.
- B. D. Dudson, N. Ben Ayed, A. Kirk, H. R. Wilson, G. Counsell, X. Xu, M. Umansky, P. B. Snyder, B. Lloyd, and the MAST Team. Experiments and simulation of edge turbulence and filaments in MAST. *Plasma Physics and Controlled Fusion*, 50(12):124012, 2008. doi: 10.1088/0741-3335/50/12/124012.
- L. Easy, F. Militello, J. Omotani, B. Dudson, E. Havlíčková, P. Tamain, V. Naulin, and A. H. Nielsen. Three dimensional simulations of plasma filaments in the scrape off layer: A comparison with models of reduced dimensionality. *Physics of Plasmas*, 21(12):122515, 2014. doi: 10.1063/1.4904207.
- Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993.
- T. Farley, N. R. Walkden, F. Militello, M. Sanna, J. Young, S. S. Silburn, J. Harrison, L. Kogan, I. Lupelli, S. S. Henderson, A. Kirk, and J. W. Bradley. Filament identification in wide-angle high speed imaging of the Mega Amp Spherical Tokamak. *Review of Scientific Instruments*, 90(9):093502, 2019. doi: 10.1063/1.5109470.
- O. E. Garcia, N. H. Bian, and W. Fundamenski. Radial interchange motions of plasma filaments. *Physics of Plasmas*, 13(8):082309, 2006. doi: 10.1063/1.2336422.
- J. D. Huba. NRL plasma formulary. Naval Research Laboratory, Washington, DC, 2019.
- Samuel Jackson, Saiful Khan, Nathan Cummings, James Hodson, Shaun de Witt, Stanislas Pamela, Rob Akers, Jeyan Thiyagalingam, and the MAST Team. FAIR-MAST: A fusion device data management system. *SoftwareX*, 27:101869, 2024. doi: 10.1016/j.softx.2024.101869.
- A. Kirk, A. J. Thornton, J. R. Harrison, F. Militello, N. R. Walkden, the MAST Team, and the EUROfusion MST1 Team. L-mode filament characteristics on MAST as a function of plasma current measured using visible imaging. *Plasma Physics and Controlled Fusion*, 58(8):085008, 2016. doi: 10.1088/0741-3335/58/8/085008.
- S. I. Krasheninnikov. On scrape off layer plasma transport. *Physics Letters A*, 283(5–6): 368–370, 2001. doi: 10.1016/S0375-9601(01)00252-3.
- H. W. Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955.
- Bruce D. Lucas and Takeo Kanade. An iterative image registration technique with an application to stereo vision. In *Proceedings of the 7th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 674–679, 1981.
- J. R. Myra, D. A. Russell, and D. A. D'Ippolito. Collisionality and magnetic geometry effects on tokamak edge turbulent transport. I. a two-region model with application to blobs. *Physics of Plasmas*, 13(11):112502, 2006. doi: 10.1063/1.2364858.
- Paola Paruta, C. Beadle, P. Ricci, and C. Theiler. Blob velocity scaling in diverted tokamaks: a comparison between theory and simulation. Technical Report WP14ER-PR(18) 21313, EUROfusion, 2018. Preprint.
- Pranab Kumar Sen. Estimates of the regression coefficient based on Kendall's tau. *Journal of the American Statistical Association*, 63(324):1379–1389, 1968.
- S. Silburn, J. Harrison, M. Smithies, A. Wynn, T. Farley, and J. Cavalier. Calcam (version 1.9.4). <https://github.com/euratom-software/calcam>, 2018.
- C. Theiler, I. Furno, P. Ricci, A. Fasoli, B. Labit, S. H. Müller, and G. Plyushchev. Cross-field motion of plasma blobs in an open magnetic field line configuration. *Physical Review Letters*, 103(6):065001, 2009. doi: 10.1103/PhysRevLett.103.065001.
-

N. R. Walkden, B. D. Dudson, L. Easy, G. Fishpool, and J. T. Omotani. Numerical investigation of isolated filament motion in a realistic tokamak geometry. *Nuclear Fusion*, 55(11): 113022, 2015. doi: 10.1088/0029-5515/55/11/113022.

S. J. Zweben, J. A. Boedo, O. Grulke, C. Hidalgo, B. LaBombard, R. J. Maqueda, P. Scarin, and J. L. Terry. Edge turbulence measurements in toroidal fusion devices. *Plasma Physics and Controlled Fusion*, 49(7):S1–S23, 2007. doi: 10.1088/0741-3335/49/7/S01.

A Camera configurations

Table 2 lists the camera configurations in the processed set, giving for each the view string from the camera metadata, the sensor column offset of the region of interest, the number of discharges, the spread of R_{LCFS} across the discharges, the Spearman correlation between limb-peak column and R_{LCFS} , the scale of the configuration where it could be fitted, and whether the configuration uses its own scale or the pooled campaign scale (Section 4.6).

Camp.	View string	ROI left	Shots	ΔR_{LCFS} (cm)	ρ	Scale (mm/px)	Used
M6	HM10, normal, f/1.8	193	70	13	-0.94	2.9 (2.7–3.0)	own
M6	HM10, normal	193	61	17	-0.89	3.7 (3.1–4.1)	own
M6	hm10 midplane f4	193	59	22	-0.47	3.9 (3.6–5.1)	pooled
M6	HM10, normal, D α 10 nm	193	30	7	-0.88	3.5 (3.2–3.9)	own
M6	HM10, normal	257	14	10	+0.34	0.9 (0.4–1.9)	pooled
M6	HM10, normal, D α 10 nm	513	7	6	–	–	pooled
M6	hm10 midplane f4	161	6	4	–	–	pooled
M6	HM10 - upside down	193	3	1	–	–	pooled
M6	probe 100kHz	193	3	6	–	–	pooled
M6	HM10, normal, D α 10 nm	545	2	0	–	–	pooled
M6	HM10, normal, D α 10 nm	225	1	0	–	–	pooled
M6	HM10, normal	225	1	0	–	–	pooled
M6	probe view	193	1	0	–	–	pooled
M7	HM10	193	22	13	-0.88	3.3 (3.0–3.8)	own
M7	HM10	257	3	1	–	–	pooled
M8	HM10 f/2.8 4.8mm nd 0.9	193	32	21	-0.58	5.2 (3.6–11.5)	own
M9	HM10 + D α filter	193	38	29	-0.82	2.7 (2.5–3.4)	own
M9	HM10	193	6	4	–	–	pooled
M9	HM10 f/2.8 4.8mm	129	4	9	–	–	pooled
M9	HM10 + D α filter	129	1	0	–	–	pooled
M9	HM10 + D α filter	513	1	0	–	–	pooled
M9	HM10 + D α filter	577	1	0	–	–	pooled
M9	HM10	129	1	0	–	–	pooled

Table 2: Camera configurations and their calibration, where a configuration uses its own scale when it has at least 8 discharges, at least 3 cm of spread in R_{LCFS} , a correlation of the right sign with $|\rho| \geq 0.5$, and a bootstrap interval that excludes zero.

B Within-shot calibration

A within-shot regression of the limb-peak column on R_{LCFS} uses the few centimetres of plasma movement during a flat top, and for the 326 discharges in which the LCFS moves by at least 2 cm, the implied scale has a median of 5.8 mm per pixel with an interquartile range of 3.8–13.6 and extremes below 1 and above 10^4 . Three effects produce this spread, since the EFIT boundary is least reliable at the start and end of the flat top, where most of the movement occurs, the brightness profile changes shape as the density rises, which moves the peak at fixed R_{LCFS} , and the lever arm is short, so that a 1 cm error in R_{LCFS} changes the slope by tens of percent. The between-shot regression of Section 4.6 replaces a few centimetres of movement in one discharge with 10–20 cm of position variation across a campaign and is correspondingly more stable.

C ELM detection

Figure 8 shows the D_α trace and the camera frame-mean brightness for an ELMy discharge with the detected spikes, and the camera proxy finds the same events as the D_α monitor where both are available and serves as the only detector where the D_α channel is missing.

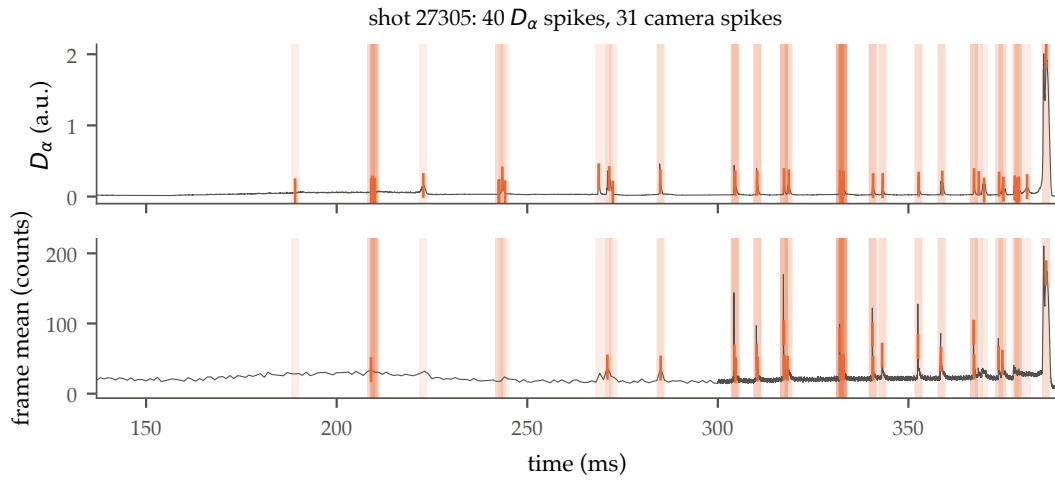


Figure 8: ELM detection on an ELMy discharge, with the midplane D_α signal and its detected spikes in the top panel, the camera frame-mean brightness and its own spike detection in the bottom panel, and the ± 1 ms exclusion windows as shaded bands.